

Precision Tissue Mapping from Spatial Transcriptomics Using Graph Learning

A neighborhood-aware framework for spatially resolved molecular pathology

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The problem. Spatial transcriptomics measures gene expression while keeping track of where each measurement sits in the tissue (a spot or a single cell, depending on the platform). This lets us ask not only which genes change in disease, but where in the tissue the change occurs, a central question in precision oncology and neuropathology [1]. Before a spatial slide can be interpreted, however, every measured spot must first be assigned to its tissue domain (a cortical layer, a tumor region, an immune niche). Today that assignment relies on slow, subjective manual annotation, or on expression-only clustering that ignores location and produces noisy, spatially incoherent maps. This project builds an automated, label-efficient model for the mapping step and tests whether it has to be spatially aware to work.

Proposed study

This project will test whether tissue-neighborhood information improves spatial-domain mapping beyond expression-only machine learning. I will use two public datasets: the LIBD DLPFC benchmark (12 Visium sections, 47,329 expert-annotated spots, 7 cortical domains) and single-cell osmFISH mouse cortex (33 genes). The main comparison will be between a spatial graph neural network and two non-graph baselines, XGBoost and an MLP, all trained on the same expression features. The key test is simple: if the graph is really helping, the GNN should perform better when it has the true tissue-neighborhood graph than when the graph is shuffled or removed.

Why this project. I chose tissue mapping because it sits where neuroscience, cancer biology, and machine learning meet, and because the question is concrete: can a model tell which part of a tissue a measurement came from? My main interest is whether local context changes the answer. A spot can be noisy on its own, but its neighbors may make its tissue domain much clearer. Testing that idea carefully is what the project is about.

1 Background and Significance

Why spatial context is clinically important. Tissue function depends on how cells are arranged in space. In the cerebral cortex, six layers (L1–L6) above the white matter each carry distinct cell types and gene programs. In a tumor, malignant cells, stroma, blood vessels, and infiltrating immune cells form spatially structured niches. Disease changes are often spatially localized: tau pathology in Alzheimer’s follows a laminar gradient, schizophrenia shows layer-specific cortical deficits, and a tumor’s response to immunotherapy depends on where immune cells sit relative to the malignant front. Standard bulk RNA sequencing averages this away. Spatial transcriptomics (10x Visium, MERFISH, Xenium, CosMx, Stereo-seq) preserves it, and a 2024 review in the Journal of Cancer Research and Clinical Oncology argues that this spatial information can advance

precision oncology by capturing tumor heterogeneity, metastasis, prognosis, and immunotherapy response that bulk profiling misses [1].

Why it matters	Consequence
Tissue function is spatially organized	Disease changes are region/layer-specific, not diffuse.
Spatial omics preserves location	We can finally ask <i>where</i> pathology starts.
But spots must be assigned to a domain first	Domain mapping is an important early step for most downstream spatial analyses.

Table 1: Accurate spatial-domain mapping is the step that turns a spatial-transcriptomics slide into interpretable, region-resolved biology, in both brain disease and cancer.

2 The Computational Gap

Domain mapping is hard for two complementary reasons, and current practice addresses neither well.

Gap 1. Manual annotation does not scale. Expert annotation of a single section is slow and subjective. A disease cohort spans dozens of sections across many donors, where hand-labelling becomes the bottleneck.

Gap 2. Per-spot expression is weak and structure-blind. A single spot’s expression is sparse and dropout-corrupted, so models that classify each spot in isolation (XGBoost, MLP) reach an accuracy ceiling and yield spatially fragmented domains.

The observation behind our approach is straightforward: a spot’s domain is largely a property of its spatial neighbourhood. Adjacent spots usually share a domain, so the signal missing from one noisy spot is present in the tissue around it (Fig. 1). A model that can use the neighbourhood should therefore do better than one that cannot, and the size of that gap is a direct measure of how much the spatial structure is worth.

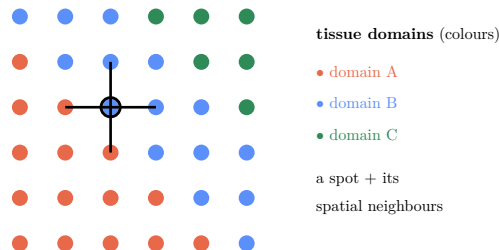


Figure 1: Domains are spatially contiguous, so a spot’s label tends to agree with its neighbours’. A single noisy spot may be ambiguous, but its neighbourhood reveals its domain, which is the signal a spatial model can use and a per-spot model cannot.

3 Central Hypothesis and Specific Aims

Central hypothesis. A spot’s tissue domain is encoded not only in its own gene expression but in its spatial neighbourhood. A graph neural network that performs tissue-neighbour message passing should therefore map domains more accurately than XGBoost and MLP models trained on the same expression features, and removing the spatial graph should erase that advantage.

Specific Aims

Aim 1. Build and benchmark. Develop a spatial GNN for tissue-domain mapping and compare it directly against tuned XGBoost and an MLP on the same features, under leakage-safe splits.

Aim 2. Test whether the advantage is spatial. Run a graph-removal test (intact vs. shuffled vs. empty graph) on two tissues and technologies, and check whether the GNN loses its advantage when the structure is destroyed.

Aim 3. Ground it biologically and extend to cancer. Recover known domain marker genes, flag candidate annotation errors, and pilot the same pipeline on tumor spatial transcriptomics to map microenvironment niches.

4 Relationship to Prior Work

GNN-based spatial-domain identification is an established field. SpaGCN [7], STAGATE [6], and GraphST [5] all apply graph neural networks to spatial-omics data and are standard baselines on the DLPFC benchmark. This proposal does not claim a new architecture. Those methods are predominantly unsupervised (graph autoencoders followed by clustering, scored by ARI against manual labels) and are compared mainly against other clustering methods. My goal is to ask a complementary and, in this field, under-asked question: how much does spatial structure actually add over strong non-graph models given the same features? I will answer it with (i) a supervised, leakage-safe cross-section protocol (train on whole sections, predict held-out ones, which is the disease-cohort regime); (ii) tuned XGBoost and MLP baselines on the same features, which spatial-domain papers rarely report; (iii) a graph-removal test that separates the spatial signal from model capacity; and (iv) replication across two technologies (Visium and osmFISH). The deliverable will be a careful estimate of the value of tissue structure. It is planned as a rigorous benchmark and analysis, not a state-of-the-art or novel-method claim.

5 Datasets

Dataset	Content	Role
LIBD DLPFC (10x Visium) [2, 3]	12 sections, 47,329 spots, expert manual layer labels (L1–L6 + WM)	Primary ground-truth benchmark (cross-section)
osmFISH mouse cortex [4]	single-cell resolution, 33 genes, ~4,839 cells, region labels	Second technology / generalization
Controlled synthetic tissue	domains defined purely by neighbourhood; per-spot signal weak by construction	Mechanism isolation / sanity check
Tumor ST (Visium/Xenium, stretch)	public solid-tumor sections with niche annotations	Clinical extension (Aim 3)

Table 2: All data are public and de-identified. LIBD DLPFC is the field-standard ground truth for spatial-domain methods. osmFISH tests a different assay with far fewer genes, where per-cell features are weakest.

6 Methodology

From tissue to spatial graph. Each section becomes a graph: nodes are spots or cells, node features are preprocessed expression (normalize, log1p, highly-variable genes, then PCA), and each node is connected to its k nearest spatial neighbours. The 2D coordinates are used only to build the graph, never as model features, so any GNN benefit is attributable to message passing over tissue structure rather than to memorizing absolute position.

6.1 Model: a spatial graph neural network

The GNN classifies each spot by pooling its neighbourhood over L rounds of message passing. With a GraphSAGE encoder,

$$h_v^{(l)} = \sigma\left(W^{(l)} \left[h_v^{(l-1)} \parallel \text{AGG}_{u \in \mathcal{N}(v)} h_u^{(l-1)} \right]\right), \quad \hat{y}_v = \text{softmax}(W_o h_v^{(L)}), \quad (1)$$

trained with cross-entropy on annotated spots. Aggregation denoises the sparse per-spot signal, which is the property Gap 2 lacks. We compare **GraphSAGE**, **GAT** (graph attention, which learns per-neighbour weights), and GCN encoders; all use PyTorch Geometric.

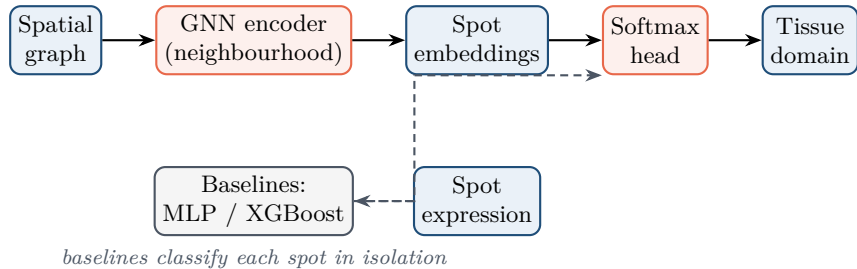


Figure 2: Pipeline. The GNN (orange) pools each spot’s spatial neighbourhood; the non-spatial baselines (gray) classify each spot alone from the same features. The only difference between them is access to the spatial graph.

6.2 Baselines and leakage-safe evaluation

I will benchmark against tuned XGBoost and an MLP on the same expression features, so the comparison isolates the value of tissue structure. I will use three splits: (i) *cross-section*, where whole sections are held out for test (layer layouts differ per section, so absolute position cannot transfer and only neighbourhood-relative reasoning generalizes; this is the disease-cohort regime and our headline test); (ii) *within-section*, where contiguous spatial blocks are held out; and (iii) *stratified*, a standard transductive node split for single-section data (osmFISH). In the cross-section disjoint-union graph, train and test sections are separate connected components, so message passing cannot leak labels across the split. I will report accuracy, macro-F1 (domains are imbalanced), and the adjusted Rand index (ARI, the field-standard domain metric), as mean $\pm$ std over seeds.

7 Planned Evaluation

The main outcome will be whether adding spatial-neighborhood information improves domain mapping compared with expression-only models. I will report accuracy, macro-F1, and adjusted Rand index (ARI). Macro-F1 is important because tissue domains are not equally sized, and ARI is commonly used in spatial-domain papers.

Question	Comparison	Evidence I will look for
Does spatial context help?	GNN vs. XGBoost / MLP	Higher macro-F1 and ARI on held-out sections or cells.
Does the graph itself matter?	True graph vs. shuffled/empty graph	Performance drops when tissue neighborhoods are destroyed.
Does the method generalize?	DLPFC Visium vs. osmFISH	Similar pattern across two spatial-omics technologies.
Is the result biologically reasonable?	Predicted domains vs. marker genes	Recovery of known layer/domain marker genes.

Table 3: Planned evaluation. The goal is not only to get a high score, but to test whether the score depends on real spatial tissue structure.

8 Rigor and Biological Interpretation

8.1 Testing whether the spatial structure is doing the work

The most important control will be a graph-removal experiment. I will train the same GNN under three graph settings: the real spatial k NN graph, a shuffled graph with similar degree but incorrect neighbors, and an empty graph with no neighbor messages. If the hypothesis is correct, the real graph should perform best, while the shuffled and empty graphs should move closer to the MLP baseline. If that does not happen, then the model’s advantage is probably not truly spatial, and I will report that directly.

8.2 From accuracy to biology

For each predicted domain I will extract the genes driving predictions and check whether known cortical-layer marker genes are recovered in DLPFC. I will also look for spots where the model confidently disagrees with the manual label. Those spots could be model errors, annotation errors, or small local tissue structures that deserve closer review. This is important because the project

should produce more than an accuracy number; it should produce biological hypotheses that a researcher can inspect.

9 Clinical Translation: Toward Precision Pathology

The cortical-layer benchmark serves as the proof of concept. The same capability, accurate and automated domain mapping that stays spatially coherent, is one requirement for translating spatial transcriptomics to the clinic, particularly in oncology [1].

Application	How domain mapping helps
Tumor microenvironment	Separate tumor core, invasive front, stroma, and immune niches to study heterogeneity and metastasis.
Immunotherapy response	Localize where immune cells sit relative to malignant cells, a spatial correlate of response.
Neuro-/neurodegenerative disease	Pinpoint which cortical layers carry early molecular pathology (Alzheimer’s, schizophrenia).
Pathologist support	Annotate complex molecular slides faster and more reproducibly than manual labelling.

Table 4: The brain benchmark tests the method; precision oncology and pathology are where it pays off. All predictions are research-only and hypothesis-generating.

10 Project Strengths and Feasibility

A few things make the project feasible and worthwhile (Table 5): it uses real, public data; it has a clear medical motivation; it compares against tuned non-graph baselines under leakage-safe splits; it includes a causal check on where the gain comes from; and it ties the accuracy back to known biology.

Criterion	Commitment
Real, sizable public data	LIBD DLPFC: 12 sections, 47k annotated spots, plus a second technology (osmFISH).
Clear medical motivation	Spatially-resolved localization of disease pathology; precision oncology bridge.
Rigorous evaluation	Tuned XGBoost and MLP baselines; leakage-safe splits; multi-seed macro-F1 and ARI.
Causal / falsification test	Graph-removal ablation will test whether the GNN depends on real tissue neighborhoods.
Interpretability → discovery	Recover marker genes; surface candidate annotation errors.
Transparent reporting	Any weak or negative result will be reported as-is.

Table 5: Strengths and feasibility of the project.

11 Eight-Week Timeline

The eight-week plan is designed to start with a reproducible baseline and then add the graph model, controls, biological interpretation, and a small tumor-ST extension if time allows (Table 6).

Weeks	Milestones
1–2	Build the data pipeline for DLPFC and osmFISH; construct spatial graphs; run a small synthetic sanity check.
3–4	Implement expression-only baselines (XGBoost, MLP) and the first GraphSAGE model with shared preprocessing.
5	Run leakage-safe evaluation splits and report macro-F1, ARI, and accuracy across seeds.
6	Run graph controls: true graph, shuffled graph, empty graph, and neighborhood-size sensitivity.
7	Biological interpretation: marker-gene recovery and confident disagreement with manual labels.
8	Prepare figures, tables, manuscript draft, and a reproducible code/data manifest; add a small tumor-ST pilot if time permits.

Table 6: Schedule for the eight-week SRI cohort. The tumor-ST extension is a stretch goal after the main brain and osmFISH benchmarks are complete.

12 Reproducibility and Ethics

Implementation will use open tools (PyTorch Geometric for the GNN; scanpy and anndata for spatial-omics I/O; XGBoost). I will save code, configs, random seeds, and a data manifest, and every reported number will be multi-seed with standard deviations. No human subjects are involved; all inputs are public, de-identified post-mortem or model-organism tissue. Predictions are research-only and hypothesis-generating and do not constitute clinical guidance.

References

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